

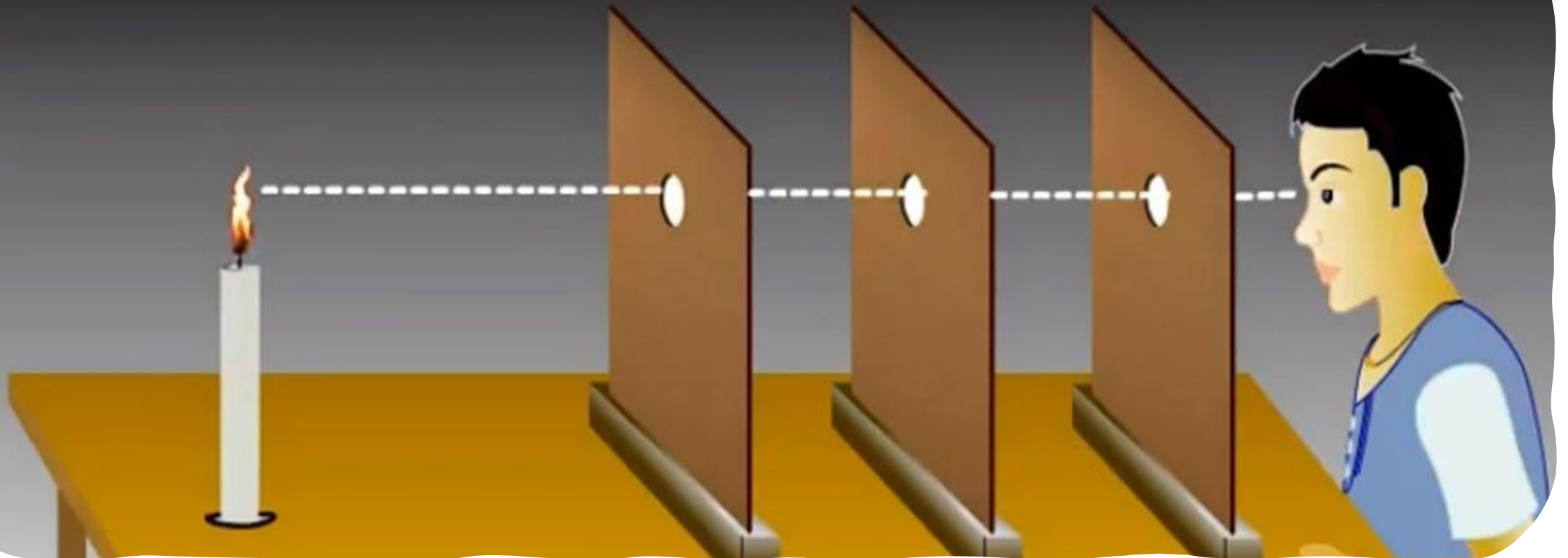
Propagation of Light

Chapter 9

PRINCIPLE OF RECTILINEAR PROPAGATION OF LIGHT

- The principle of rectilinear propagation of light states that **light travels in straight lines**.
- This means that light travels in a straight line unless it is reflected, refracted, or absorbed by an object.

Rectilinear Propagation of Light

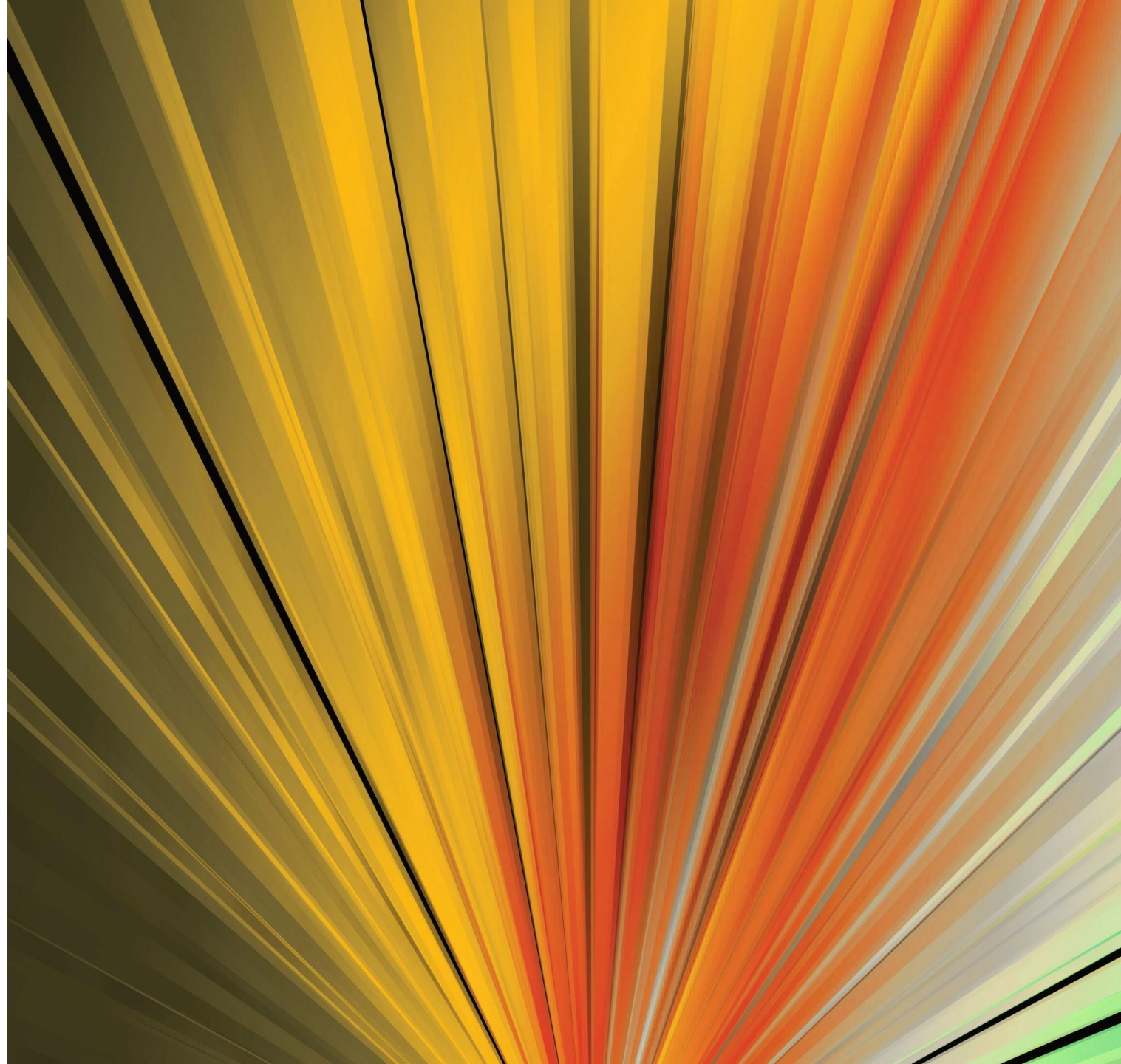


Why Does It Matter?

- When you look at a book, you see the words on the page because light from the words travels in a straight line from the book to your eyes. If there were an object in the way, you would not be able to see the words.
- In many optical instruments, such as telescopes and microscopes, lenses and mirrors are used to bend light rays so that they can be focused on a particular point. This allows us to see objects that are far away or very small.
- The principle of rectilinear propagation of light is a fundamental principle of optics. It is used to explain many different optical phenomena, and it is essential for the design and construction of optical instruments.

LIGHT BEAMS

- A light beam is a collection or bundle of light rays that are traveling in a certain direction.
- Light rays are lines that represent the direction of light travel. They are drawn as straight lines, even though light is actually a wave.



Light Beams Can Be

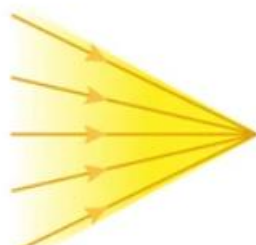
Parallel	Divergent	Convergent
light rays are all traveling in the same direction and are equidistant from each other. This type of beam is often produced by a laser.	light rays are all traveling away from a common point. This type of beam is often produced by a flashlight or a candle.	light rays are all traveling towards a common point. This type of beam is often produced by a magnifying glass or a telescope.



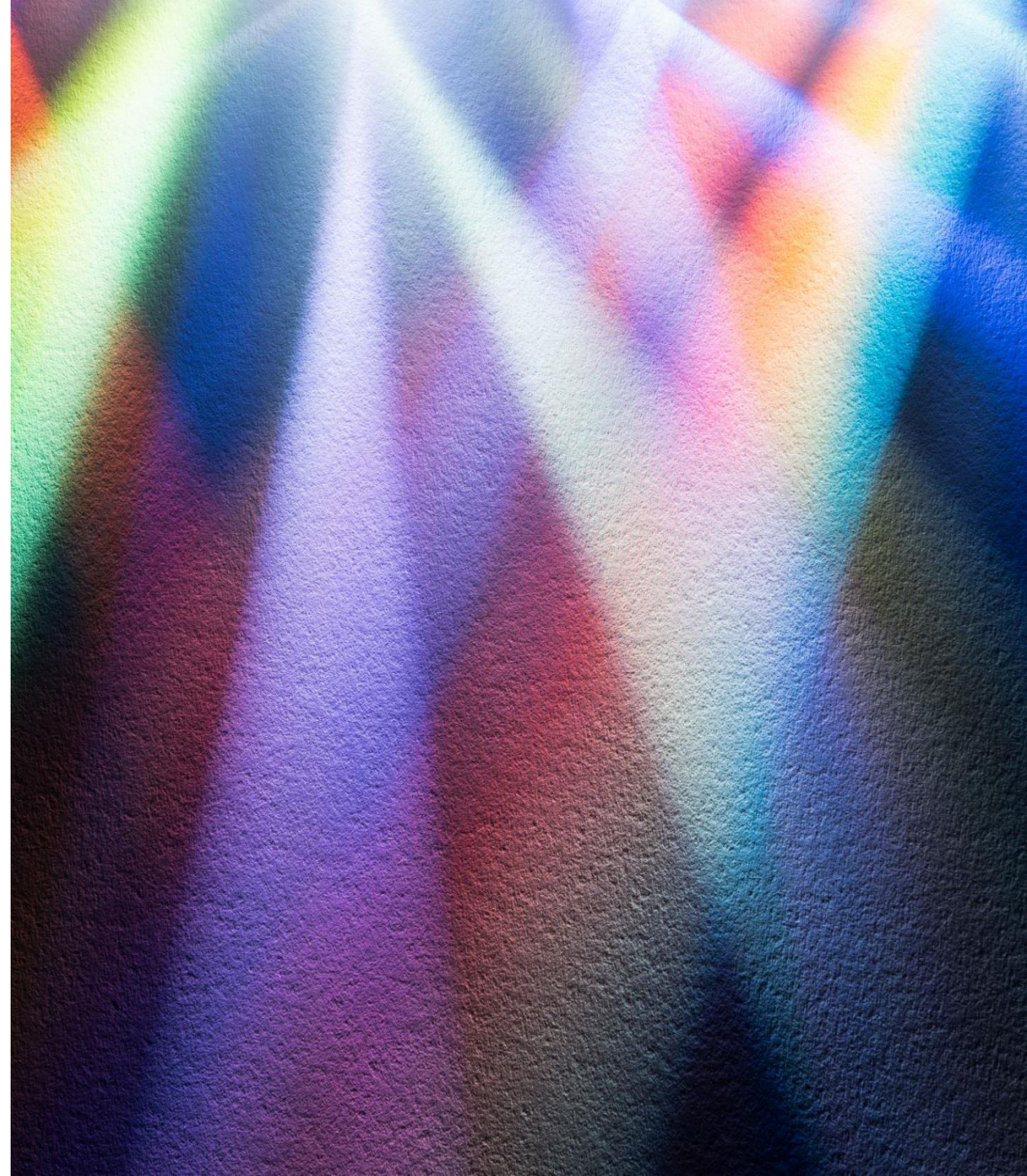
Parallel Beam



Diverging Beam



Converging Beam



OBJECTS AND IMAGES



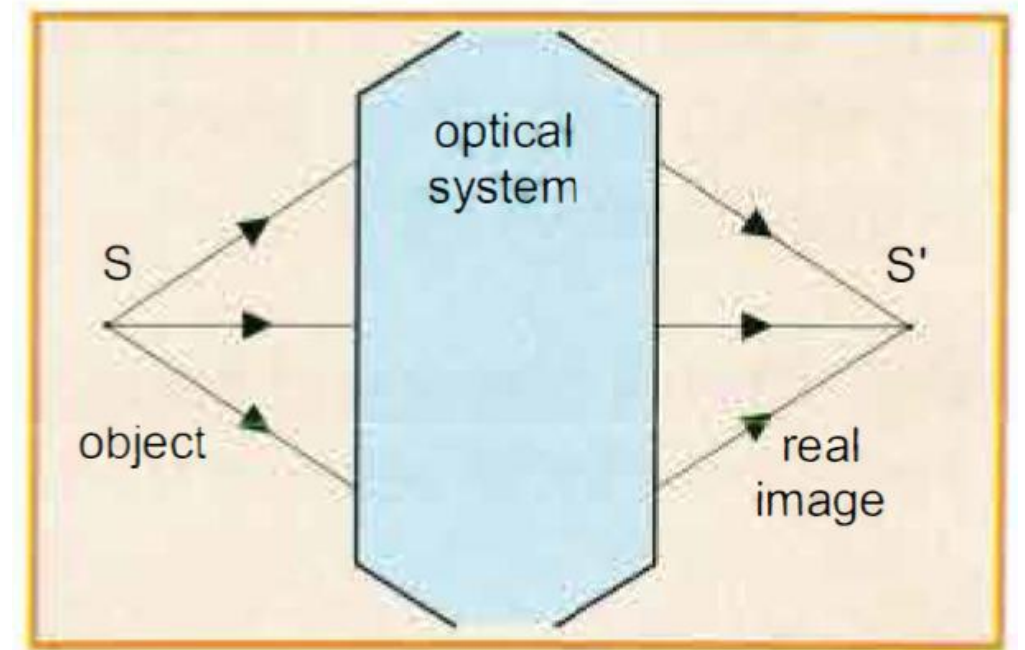
In optics, an object is any **source of light** falling on an optical system. The Sun, a candle, a lamp, etc., are examples of **objects**.



The **Moon**, an **illuminated** wall, etc., **reflect light** falling on them and are also considered objects.

OBJECTS AND IMAGES

- If you stand in front of a plane mirror, you see your "**image**" through the mirror. This image is of the same shape and size as you are.
- If you look at a building through the "eye" of a camera, you also see the image of the building. This image is not of the same size as the building.
- In these cases, what you see is called the image of the object given by the **optical system** (the mirror or the lens of the camera).
- In general, an optical system forms an image of any object placed in front of it.



REAL IMAGES VS VIRTUAL IMAGES



A **real image** can be collected on a screen. It is formed by a converging beam emerging from the optical system.



A **virtual image** cannot be collected on a screen, like your image in the plane mirror. The rays emerging from the optical system (the mirror) form a divergent beam, which appears to come from the virtual image.



The eye can see both real and virtual images. In all cases, if B' is the image of a point object B , any ray issued from B toward the system emerges from it and **passes through B' if B' is real**, or **seems to come from B' if B' is virtual**.

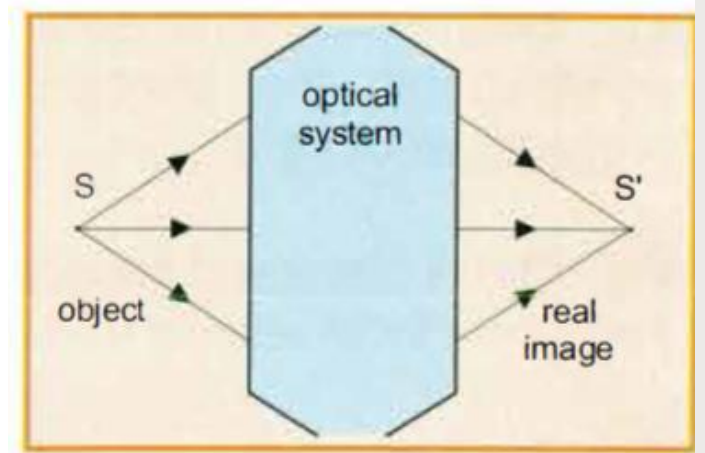


Figure 9.10. A real image is formed by a converging beam emerging from an optical system.

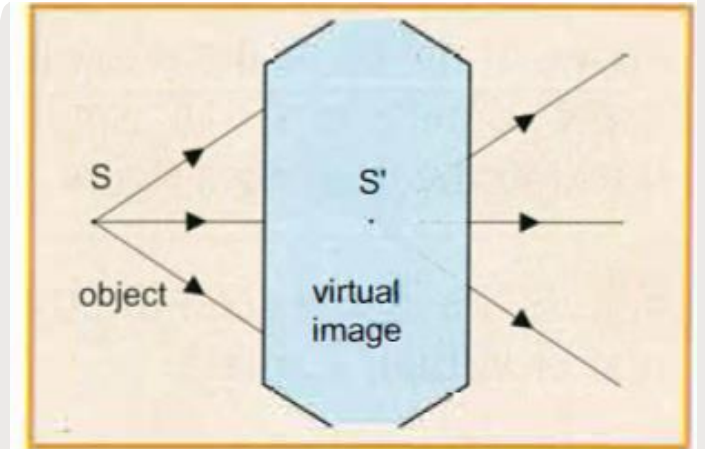


Figure 9.11. A virtual image appears to issue a divergent beam emerging from the optical system.

Summary

- * Light propagates in straight lines.
- * Beams of light can be converging, diverging, or parallel.
- * Real images are formed by converging beams of light emerging from optical systems and can be collected on a screen.
- * Virtual images appear to issue divergent beams of light emerging from optical systems and cannot be collected on a screen.